

Chapter - 3



PLANT KINGDOM



NEET
Focus

→ Diversity of Plants and Their Classification

1. INTRODUCTION

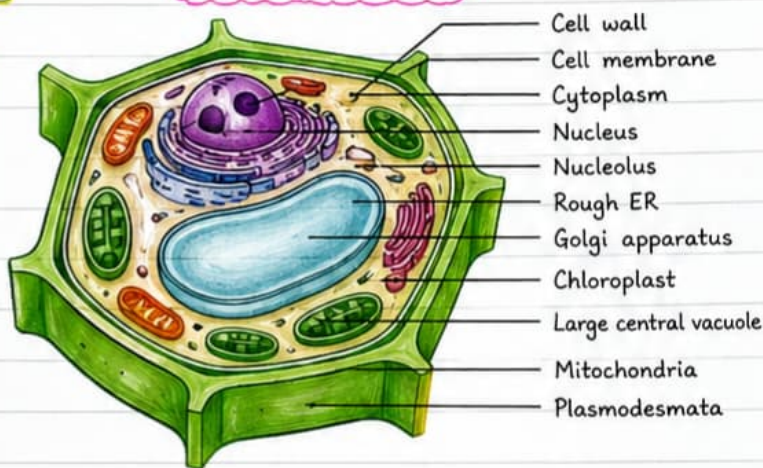
- Plants are **multicellular**, **eukaryotic** and **photosynthetic** organisms.
- They form the base of almost all ecosystems.
- Plants are also called '**Producers**' as they prepare their own food.
- They are essential for life on Earth as they release **oxygen** and maintain **ecological balance**.



2. GENERAL CHARACTERS OF PLANTS

- Eukaryotic and **multicellular**
- Cell wall made up of **cellulose**
- Have plastids containing **chlorophyll**
- Store food in the form of **starch**
- Generally non-motile
- Exhibit growth throughout life
- Show alternation of generations in life cycle

TYPICAL PLANT CELL



3. BASIS OF CLASSIFICATION

- **Cryptogams** (Non-flowering plants) → Do not produce seeds
- **Phanerogams** (Flowering plants) → Produce seeds

4A. CRYPTOGRAMS (Non-flowering Plants)

- ① Thallophyta → **Algae**
- ② Bryophyta → **Mosses, Liverworts**
- ③ Pteridophyta → **Ferns**

Key Features:

- No true roots, stems, leaves
- Reproduce by spores
- Require water for fertilization
- Mostly found in moist habitats



4B. PHANEROGAMS (Flowering Plants)

- ① Gymnosperms → **Naked seeds**
- ② Angiosperms → **Seeds enclosed in fruit**



- Flower** → Reproductive part
- Leaf** → Site of photosynthesis
- Stem** → Supports the plant
- Root** → Absorbs water and minerals

5. IMPORTANCE OF PLANTS

- ✓ Provide oxygen and food
- ✓ Maintain ecological balance
- ✓ Used in medicines, timber, fibres
- ✓ Support life directly or indirectly



QUICK FACTS

- ★ Algae → Simplest plant
- ★ Mosses → Amphibians of plant kingdom
- ★ Ferns → First true vascular plants
- ★ Angiosperms → Most advanced



CHAPTER-3 : PLANT KINGDOM

6. ALGAE – Plant Body Thallus

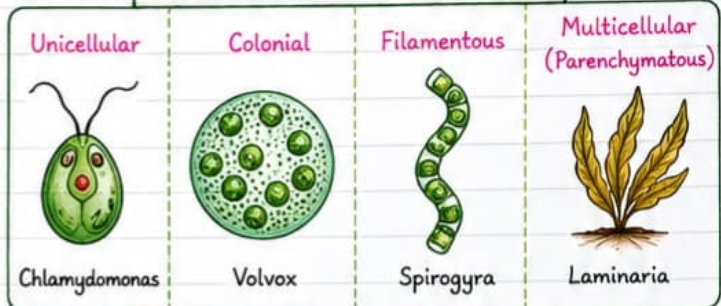
- Mostly **aquatic** (freshwater & marine).
- Thallus is **simple** (unicellular, colonial, filamentous or multicellular).
- Cell wall is made of **cellulose**.
- Have **chlorophyll a** and other pigments.
- Food is stored as **starch**.
- Reproduce by **zoospores**, **aplanospores**, or by fragmentations.

Examples

Chlamydomonas, Volvox, Spirogyra, Ulothrix, Diatoms, Laminaria.



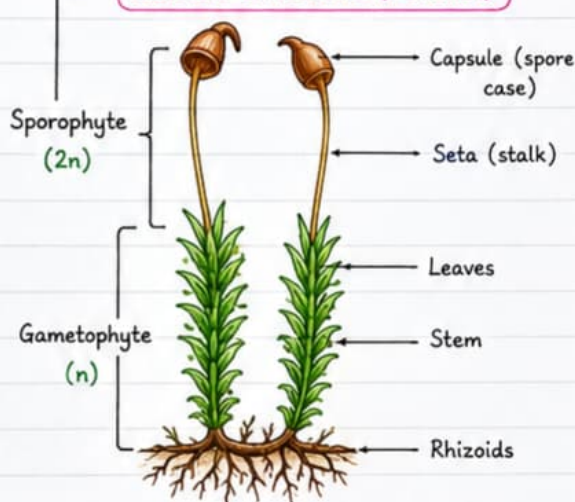
TYPES OF THALLUS IN ALGAE



7. BRYOPHYTA – Amphibians of Plant Kingdom

- Live in **moist & shady** places.
- Plant body is differentiated into leaf-like, stem-like and root-like structures called **rhizoids**.
- Vascular tissue is **absent**.
- **Gametophyte** is **dominant**.
- Sex organs are **multicellular** and **jacketed**.
- Require water for fertilization.
- **Examples**: Marchantia (Liverwort), Funaria (Moss).

GENERAL STRUCTURE (FUNARIA)



Life Cycle

Gametophyte (n) produces gametes
 → fertilization (in water) → **Zygote (2n)** → **Sporophyte (2n)** → produces spores (n) by meiosis → new gametophyte.

8. PTERIDOPHYTA – First True Vascular Plants

- ★ Have well developed roots, stems & leaves.
- ★ Vascular tissues (xylem & phloem) are present.
- ★ Plant body is **sporophyte**.
- ★ Spore producing organs are called **sporangia**, borne on sporophylls.
- ★ Sporangia are grouped in **sori**.
- ★ Require water for fertilization.
- ★ **Examples**: Lycopodium, Selaginella, Equisetum, Pteris, Dryopteris.

FERN (PTERIS)



SORUS (on lower side of frond)



SIGNIFICANCE OF PTERIDOPHYTES

- ✓ Help in soil formation.
- ✓ Some are ornamental plants.
- ✓ Useful in biological studies.
- ✓ Indicate the beginning of vascular system in plants.



QUICK RECAP

- **Algae** → Simple, mostly aquatic, no true roots, stems, leaves.
- **Bryophyta** → Non-vascular, need water for fertilization.
- **Pteridophyta** → Vascular, spore bearing, first true land plants.



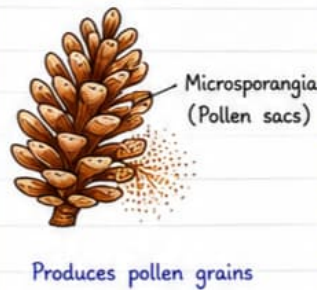
CHAPTER-3 : PLANT KINGDOM

9. GYMNOSPERMS – Naked Seed Plants

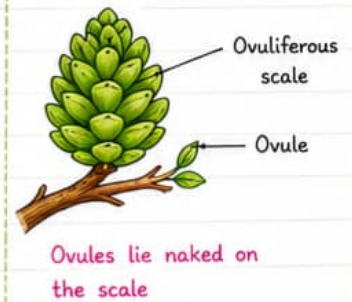
- Seeds are 'naked' (not enclosed in fruit).
- Reproductive structures are **cones** (strobili).
- Xylem usually consists of **tracheids** (no vessels).
- Mostly woody trees or shrubs.
- **Examples** : Cycas, Pinus, Cedrus, Ginkgo.

CONES (STROBILI)

MALE CONE (Microstrobilus)



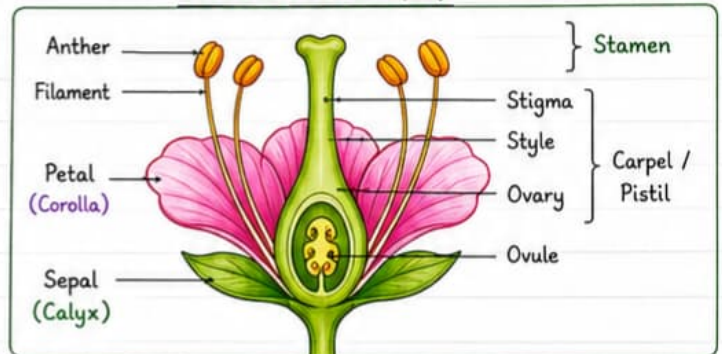
FEMALE CONE (Megastrobilus)



10. ANGIOSPERMS – Flowering Plants

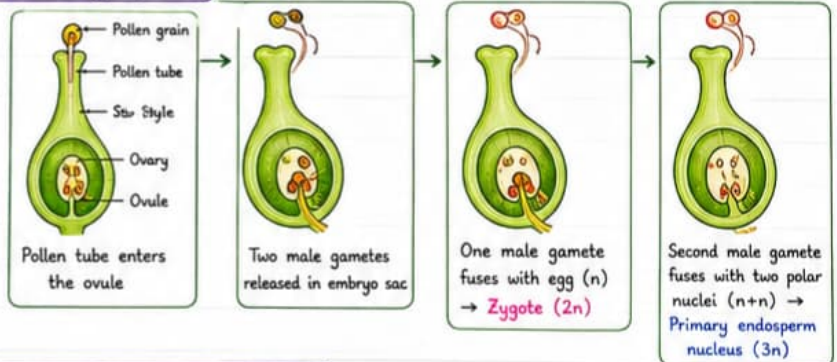
- Most advanced and largest group of plants.
- Seeds are **enclosed** within fruits.
- Have well developed **vascular** tissues.
- Exhibit **double fertilization**.
- Show great diversity in habit and habitat.
- **Examples** : Mango, Maize, Pea, Rose, Sunflower.

TYPICAL FLOWER (LS)



11. DOUBLE FERTILIZATION – Unique in Angiosperms

- One male gamete fuses with egg cell → forms **Zygote** ($2n$).
- Second male gamete fuses with two polar nuclei → forms **Primary endosperm nucleus** ($3n$).
- Endosperm nourishes the developing embryo.



12. COMPARISON – BRYOPHYTES, PTERIDOPHYTES & SEED PLANTS

Feature	Bryophytes	Pteridophytes	Seed Plants	
			Gymnosperms	Angiosperms
Vascular tissue	Absent	Present	Present	Present
True roots, stems, leaves	Absent	Present	Present	Present
Reproductive organs	Antheridia, Archegonia	Antheridia, Archegonia	Cones (Strobili)	Flowers
Seeds	Absent	Absent	Naked seeds	Seeds enclosed in fruits
Examples	Marchantia, Funaria	Pteris, Fern, Equisetum	Pinus, Cycas, Cedrus	Mango, Pea, Sunflower

Key Takeaway

Plant kingdom shows a clear evolutionary progression from simple, non-vascular plants to complex vascular plants with seeds and flowers.



IMPORTANT POINTS

- ✓ Plants are essential for life on Earth.
- ✓ Algae → Simple, mostly aquatic.
- ✓ Bryophytes → First amphibians of plant kingdom.
- ✓ Pteridophytes → First true vascular plants.
- ✓ Gymnosperms → Naked seeds.
- ✓ Angiosperms → Seeds enclosed in fruits.

QUICK REVISION

- ★ Algae – Thallus body, no true roots/stems/leaves.
- ★ Bryophytes → Require water for fertilization.
- ★ Pteridophytes → Sporangia in sori.
- ★ Gymnosperms → Cones are reproductive organs.
- ★ Angiosperms – Flowers, fruits & double fertilization.

CHAPTER-3 PLANT KINGDOM

Summary & Important Points

13. ECONOMIC IMPORTANCE OF PLANTS

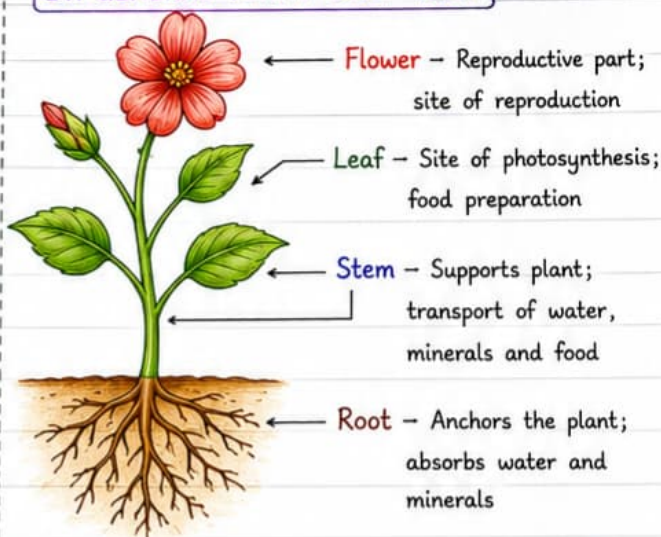
- Food** – Cereals, pulses, fruits, vegetables, spices.
- Oxygen** – Maintain balance of gases in atmosphere.
- Medicines** – Source of many important drugs (e.g., Cinchona, Rauwolfia).
- Raw Materials** – Wood, fibres, rubber, gums, resins, oils.
- Ecosystem Services** – Prevent soil erosion, maintain water cycle, support wildlife.

Examples of Useful Plants

- Neem – Antiseptic
- Tulsi – Immunity booster
- Cinchona – Source of Quinine
- Rubber plant – Source of rubber
- Sunflower – Edible oil



14. GENERAL PLANT BODY PLAN



All plant parts work **together** for survival and reproduction.

15. ALGAE, BRYOPHYTES, PTERIDOPHYTES, GYMNASPERMS & ANGIOSPERMS – AT A GLANCE

Group	Habitat	Body Structure	Reproductive Structure	Example
Algae	Aquatic	Thallus	Spores, aplanospores, zoospores	Chlamydomonas, Spirogyra
Bryophytes	Moist, shady	Rhizoids, leaf-like, stem-like	Spores in capsule	Marchantia, Funaria
Pteridophytes	Moist soil	True roots, stems, leaves	Spores in sporangia (sori)	Pteris, Equisetum
Gymnosperms	Terrestrial	Well developed vascular tissues	Naked seeds in cones	Pinus, Cycas
Angiosperms	Terrestrial	Well developed vascular tissues	Seeds enclosed in fruits; flowers	Mango, Sunflower

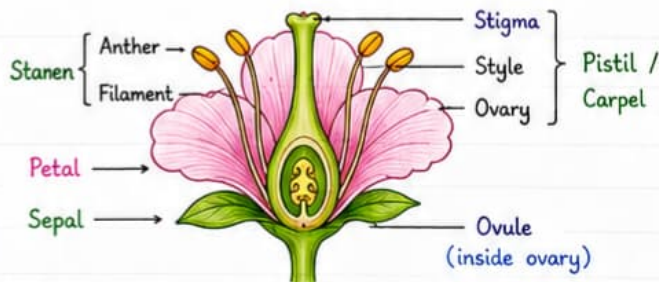
16. KEY FACTS TO REMEMBER

- ★ Algae are the simplest plants.
- ★ Bryophytes are the first amphibians of plant kingdom.
- ★ Pteridophytes are the first true vascular plants.
- ★ Gymnosperms have naked seeds.
- ★ Angiosperms are most advanced and dominant group.
- ★ **Vascular tissue** (xylem & phloem) is absent in Algae and Bryophytes.
- ★ Double fertilization occurs only in Angiosperms.

Remember This!

Plants are the **green factories** of Earth.
They make **life possible** for all living beings.

17. FLOWERING PLANT – PARTS OF A FLOWER



SHORT QUESTIONS (For Quick Practice)

- What are the general characteristics of plants?
- Differentiate between Bryophytes and Pteridophytes.
- What is double fertilization? In which plants does it occur?
- Write short note on economic importance of plants.
- Draw a well-labelled diagram of L.S. of a flower.